ARANI ROY

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EDUCATION

Ph.D., Electrical and Computer Engineering, GPA: 3.8/4.0 — Aug 2021 – present
Purdue University, West Lafayette, IN
Advisor: Dr. Kaushik Roy
Relevant Coursework: Deep Learning, Artificial Intelligence, Optimization in Deep Learning, Linear Algebra, Data Mining, AI Hardware, Advanced VLSI

B.E., Electronics and Telecommunication Engineering, GPA: 9.35/10.0 — 2014 – 2018
Jadavpur University, Kolkata, India

RESEARCH INTERESTS

Multimodal Generative AI, Diffusion Models, Controllable Generation, Concept Erasure/Safety, Model Compression

RESEARCH EXPERIENCE & PROJECTS

Graduate Research Assistant — Aug 2021 – Present
Purdue University, Center for Brain-Inspired Computing, West Lafayette, IN

Completed Projects:

- **Stable Diffusion Concept Control (NeurIPS 2026 under review):** Proposed a geometry-aware framework for controllable diffusion generation using hyperspherical traversal in text embedding space, enabling fine-grained subject and attribute control across UNet and DiT architectures.
- **Diffusion model compression (under review at ECCV 2026):** Pioneered a closed-form, activation-guided rank-reduction pipeline for attention matrices, which is training-free and timestep-aware under a user-specified compression budget, achieving **35% inference acceleration** and **∼100M parameter reduction**. Additionally, provided theoretical guarantees of bounded pruning error and reduced calibration overhead by 98% via “SlimSet,” a 500-sample representative coreset distilled from 30k images.
- **Backpropagation-free networks (WACV 2026 Oral, CVPRW 2025):** Proposed a structured local learning framework operating on SVD manifolds that scales Direct Feedback Alignment to **32-layer** ResNets on ImageNet, and works on convolutional layers. Achieved **BP-level accuracy** via orthogonality-preserving updates on the Stiefel manifold, while demonstrating the **low inference-time compute** among biological baselines.
- **Diffusion model concept unlearning (NeurIPS 2025 Spotlight):** Designed a fast, training-free, closed-form update on cross-attention to suppress or remove target concepts in diffusion models using **orthogonal projections and spectral geometry**, operating directly in the CLIP–attention interface. Achieved **16%** improvement over training-based and free methods in removing targeted concepts, with verified resistance to adversarial red-teaming attacks.
- **Hardware-algorithm co-design (ISCAS 2025):** Developed a non-uniform, approximate-multiplier–aware quantization flow co-optimized with the target accelerator, demonstrating **3.5 x energy savings** and **1.3 x speedup**.

Ongoing Projects:

- **Multimodal Model Concept Erasure:** Revisiting the non-linear geometry in the embedding layers to target removal of harmful concepts in multimodal models like VLMs, Diffusion models etc.

Systems Intern, Kilby Labs — May 2023 – Aug 2023
Texas Instruments, Dallas, Texas

- **Compute-in-Flash for Matrix–Vector Workloads** Explored digital compute-in-flash on 28nm flash for neural-style MVMs; compared against 65nm baseline. Analyzed peripheral/sense-amp constraints for reliable in-memory MAC.

PUBLICATIONS

- **”HEART: Hyperspherical Embedding Alignment via Kent-Representation Traversal in Diffusion Models”**, Under Review at NeurIPS 2026. Authors: Arani Roy, S.D. Biswas, Kaushik Roy
- **“SlimDiff: Training-Free, Activation-Guided Slimming of Diffusion Models”**, Under Review at ECCV 2026. Authors: Arani Roy*, S.D. Biswas*, Kaushik Roy

- **“Feedback Alignment Meets Low-Rank Manifolds: A Structured Recipe for Local Learning,”** WACV 2026, **Oral, Best Student Paper-Algorithms**. Authors: Arani Roy, M. P.E. Apolinario, S.D. Biswas, Kaushik Roy
- **“CURE: Concept Unlearning via Orthogonal Representation Editing in Diffusion Models,”** NeurIPS 2025, **Spotlight**. Authors: Arani Roy*, S.D. Biswas*, Kaushik Roy
- **“AlphaBlend: Hardware-Algorithm Co-design with Mixed-Alphabet Set Multipliers for DNN Workloads”** ISCAS 2025. Authors: Arani Roy, Kaushik Roy
- **”Now You See It, Now You Don’t - Instant Concept Erasure for Safe Text-to-Image and Video Generation”**, CVPR Findings 2026. Authors: S. D. Biswas, Arani Roy, Kaushik Roy
- **“LLS: Local Learning Rule for Deep Neural Networks Inspired by Neural Activity Synchronization,”** WACV 2025. Authors: M.P. Apolinario, Arani Roy, Kaushik Roy
- **“Towards Scalable Modeling of Compressed Videos for Efficient Action Recognition”** WACV 2026. Authors: S.D. Biswas, E. Soufleri, Arani Roy, Kaushik Roy

Patents (Samsung) from Industry Experience

- TSPC edge-triggered precharge-based flip-flop.
- Performance-optimized flip-flop with negative setup time.
- True single-phase clock NAND-based reset flip-flop.

INDUSTRY EXPERIENCE

Associate Staff Engineer Jul 2018 – Jul 2021
Samsung Semiconductors India R&D, Bengaluru, India

- Led end-to-end library design projects across 4nm-130nm technology nodes, automated design workflows and achieved $\sim$ 2× faster turnaround, achieved 3 patents from performance, power and area optimized flip-flop design kits.

Compute-in-Flash Intern May 2023 – Aug 2023
Texas Instruments, (28nm flash node)

- Prototyped and evaluated a digital compute-in-flash architecture for DNN-style matrix–vector workloads, including porting a 65nm design to 28nm flash and analyzing throughput–energy trade-offs.

Student Trainee May 2017 – Jul 2017
Samsung Electronics, Bengaluru, India

- Built a synthesizable SRAM/memory compiler with RTL-to-GDS flow.

HONORS & AWARDS

- WACV 2026 **Oral**, **Award: Best Student Paper - Algorithms**
- NeurIPS 2025 **Spotlight**
- Bilsland Dissertation Fellowship, Purdue University (Fall 2026 semester of PhD)
- Frederick N. Andrews Fellowship, Purdue University (4 years of PhD)
- Samsung President’s Award (innovation in 4nm/5nm node)
- Samsung Best Intern Project (Hardware)
- Best Paper (User Design Track), 33rd International. Conf. on VLSI Design & Embedded Systems (Samsung Foundry)
- Sanghamitra Gold Medal (Bachelor’s), Jadavpur University

SKILLS

Research Areas	Diffusion Models, Multimodal Generative AI, Concept Erasure, Model Compression
Frameworks	PyTorch, Hugging Face Diffusers, CUDA
Programming	Python, Bash, C++, Java
Hardware/Systems	Verilog, HSPICE, Cadence Virtuoso, Synopsys ICC2, FineSIM

REVIEWER SERVICE

ECCV 2026, ICML 2026, CVPR 2026, ICLR 2026, ICLR 2025, WACV 2026, ICCV 2025